Generic Degree Two in $JC(2)$: Secant Projectors and Galois Rigidity

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Abstract

We study planar polynomial maps of generic degree two through the scheme of pairs with the same image. For

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}},$$

put $A = \mathbb{C}[x_1, x_2]$, let $I_R(F)$ be the collision ideal in the pair ring $S = \mathbb{C}[x_1, x_2, y_1, y_2]$, and let I_Δ be the diagonal ideal. Set $C_F = S/I_R(F)$. Their quotient

$$\mathrm{Obs}(F) = I_\Delta / I_R(F) \subseteq C_F$$

is the ideal of the diagonal in the collision ring. When F is Keller, an ordered divided-difference matrix produces an explicit idempotent $p_F \in C_F$ with $\mathrm{Obs}(F) = C_F p_F$.

Writing $B = \mathbb{C}[P, Q] \subseteq A$, $K = \mathrm{Frac}(B)$, and $L = \mathrm{Frac}(A)$, suppose that the induced extension L/K has degree two. Then the generic collision algebra decomposes compatibly with the diagonal as

$$K \otimes_B C_F \simeq_K L \times L,$$

where diagonal evaluation is first projection. The second factor forces $\mathrm{Obs}(F) \neq 0$; when F is Keller, the secant identity $\mathrm{Obs}(F) = C_F p_F$ then gives $p_F \neq 0$. On the other hand, a separable quadratic extension is Galois, and classical Keller–Galois rigidity makes a hypothetical generic-degree-two Keller map a polynomial automorphism. Then $\mathrm{Obs}(F) = 0$ and $p_F = 0$, a contradiction. Thus no planar Keller map has generic degree two. The collision and secant arguments are formalized in Lean; the classical rigidity theorem is isolated there as the sole external mathematical input to the degree-two exclusion.

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1 Introduction

The complex two-dimensional Jacobian conjecture $JC(2)$ asks whether every polynomial map

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^{\times},$$

has a polynomial inverse. Such an F is called a *Keller map*. The condition on the Jacobian is local, while polynomial invertibility is global. Collision geometry makes the global issue visible by retaining pairs (x, y) for which $F(x) = F(y)$.

The companion paper [vdV26] develops the dimension-independent collision construction. It associates to a polynomial map two ideals

$$I_R(F) = (F_i(x) - F_i(y)), \qquad I_\Delta = (x_i - y_i)$$

inside the coordinate ring of ordered pairs. The containment $I_R(F) \subseteq I_\Delta$ induces the obstruction ideal

$$\mathrm{Obs}(F) = I_\Delta / I_R(F)$$

in the collision ring C_F. This ideal defines the diagonal inside the collision scheme. Its vanishing says that the collision scheme consists only of the diagonal; over $\mathbb{C}$, this is equivalent to polynomial automorphy.

The same construction has a generic field-theoretic form. Write

$$A = \mathbb{C}[x_1, x_2], \qquad B = \mathbb{C}[P, Q] \subseteq A, \qquad K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A).$$

For a Keller map, L/K is finite and separable, and

$$d_F = [L : K]$$

is the generic degree of F. At the generic point, Paper I gives the compatible isomorphism

$$K \otimes_B C_F \simeq_K L \otimes_K L,$$

with base-changed diagonal evaluation equal to tensor multiplication.

Paper I already uses classical Galois rigidity to exclude counterexamples of generic degrees one and two in every dimension. The purpose here is to realize the quadratic exclusion explicitly in the planar collision ring. A recent preprint states the stronger exclusion of every prime generic degree [Mos24]. We do not use that broader argument; the focus here is the explicit collision certificate and its formal verification.

This paper specializes that construction to $d_F = 2$. There are two complementary descriptions of the collision obstruction. The first is geometric and explicit: ordered divided differences produce the planar multivariate Bézoutian determinant and an idempotent p_F generating $\mathrm{Obs}(F)$. The second is generic and field-theoretic: a separable quadratic extension satisfies

$$L \otimes_K L \simeq_L L \times L,$$

and tensor multiplication is projection onto the marked first factor. For a Keller map, the second factor is a genuine residual generic collision, forcing $p_F \neq 0$.

Classical Galois rigidity gives the opposite conclusion. Since every separable quadratic extension is normal, a hypothetical planar Keller map with $d_F = 2$ would be a polynomial automorphism [Cam73, Raz79, Wri81]. Its collision scheme would then be the diagonal, so $p_F = 0$. The contradiction excludes generic degree two.

Section 2 recalls, without reproving, the part of the general construction needed here. Section 3 constructs the planar secant projector. Section 4 proves quadratic collision nonvanishing, and Section 5 combines it with Galois rigidity. Section 6 characterizes vanishing through the marked normal closure and source regularity of the fixed algebra. Section 7 records the formalization boundary.

This is a generic-degree-two theorem, not a proof of all of $JC(2)$. The remaining possible counterexamples have nonnormal function-field extensions of generic degree at least three. No normalization, boundary, or inertia construction enters the degree-two proof. The separate boundary analysis below does not assume a generic-degree bound.

2 Inputs from Paper I

We use the dimension-independent construction of Paper I [vdV26]. To keep the two manuscripts notationally identical, let

$$F = (F_1, \dots, F_n) \colon \mathbb{A}^n_{\mathbb{C}} \longrightarrow \mathbb{A}^n_{\mathbb{C}},$$

and put

$$A = \mathbb{C}[x_1, \dots, x_n], \qquad S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n],$$
$$I_R(F) = (F_i(x) - F_i(y))_{i=1}^n, \qquad I_\Delta = (x_i - y_i)_{i=1}^n,$$
$$C_F = S/I_R(F), \qquad R_F = \operatorname{Spec} C_F.$$

Since $I_R(F) \subseteq I_\Delta$, diagonal substitution descends along the quotient map $q_F \colon S \twoheadrightarrow C_F$ to

$$\bar{\mu}_F \colon C_F \twoheadrightarrow A, \qquad [s] \longmapsto s(x, x).$$

The image of I_Δ is the ideal

$$\mathrm{Obs}(F) := q_F(I_\Delta) = I_\Delta / I_R(F) \subseteq C_F,$$

which defines the diagonal inside R_F.

The following two results are quoted from Paper I; no proof of the general collision construction is repeated here.

Theorem 2.1 (Diagonal kernel and automorphism detection)**.** *For every complex polynomial map as above,*

$$\ker(\bar{\mu}_F) = \mathrm{Obs}(F) = I_\Delta / I_R(F).$$

Moreover,

$$F \text{ is a polynomial automorphism} \iff I_R(F) = I_\Delta \iff \mathrm{Obs}(F) = 0.$$

The implication $\mathrm{Obs}(F) = 0 \Rightarrow F$ is a polynomial automorphism uses Ax–Grothendieck [Ax69]. The degree-two exclusion needs only the elementary forward implication.

Let

$$B = \mathbb{C}[F_1, \dots, F_n] \subseteq A, \qquad K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A),$$

and let

$$\mu_L \colon L \otimes_K L \longrightarrow L, \qquad \ell \otimes \ell' \longmapsto \ell\ell'$$

be tensor multiplication.

Theorem 2.2 (Generic collision bridge)**.** *If L/K is finite, then*

$$K \otimes_B A \simeq_K L$$

and there is a K-algebra isomorphism

$$\Phi_F \colon K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L$$

which carries the base-changed diagonal evaluation to μ_L. Furthermore, flatness of the localization $B \to K$ identifies

$$K \otimes_B \mathrm{Obs}(F) \simeq \ker(1 \otimes \bar{\mu}_F)$$

as K-vector spaces.

These statements combine the fiber-product, generic-base-change, and diagonal-kernel results of Paper I. Their compatibility with μ_L marks which generic factor is the diagonal.

Return now to

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}.$$

For a Keller map, P, Q are algebraically independent and L/K is finite separable. We write

$$d_F := [L : K]$$

for its generic degree. The 2 in $JC(2)$ denotes ambient dimension; the separate condition $d_F = 2$ concerns the generic fiber. The nonvanishing argument below only uses a finite separable extension of degree two. The Keller condition enters through the nonzero constant Jacobian and, at the end, through Galois rigidity.

3 The explicit planar secant projector

The matrix below is the ordered multivariate Bézoutian matrix associated to F; compare [McK23, SS75]. We retain the term *secant matrix* because its determinant separates the diagonal collision sheet from its complement.

Let

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF = c \in \mathbb{C}^\times.$$

Set

$$S = \mathbb{C}[x_1, x_2, y_1, y_2], \qquad \mathbf{d} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \end{pmatrix}, \qquad \mathbf{r} = \begin{pmatrix} P(x) - P(y) \\ Q(x) - Q(y) \end{pmatrix}.$$

3.1 Ordered divided differences

For $H \in \mathbb{C}[x_1, x_2]$, define

$$D_1 H = \frac{H(x_1, x_2) - H(y_1, x_2)}{x_1 - y_1},$$
$$D_2 H = \frac{H(y_1, x_2) - H(y_1, y_2)}{x_2 - y_2}.$$

Both quotients lie in S. The ordered telescoping identity is

$$H(x) - H(y) = D_1 H\,(x_1 - y_1) + D_2 H\,(x_2 - y_2).$$

Define

$$M_F = \begin{pmatrix} D_1 P & D_2 P \\ D_1 Q & D_2 Q \end{pmatrix}.$$

Then

$$\mathbf{r} = M_F \mathbf{d}, \qquad M_F\big|_\Delta = JF.$$

Let

$$\delta_F := \det(M_F) \in S.$$

Proposition 3.1 (Secant annihilation)**.** *The secant determinant satisfies*

$$\delta_F I_\Delta \subseteq I_R(F), \qquad \delta_F\big|_\Delta = c.$$

Proof. Multiplying $\mathbf{r} = M_F \mathbf{d}$ by the adjugate matrix gives

$$\delta_F \mathbf{d} = \operatorname{adj}(M_F)\mathbf{r}.$$

Both entries on the right belong to $I_R(F)$. Since the entries of $\mathbf{d}$ generate I_Δ, the first assertion follows. Restriction to the diagonal gives

$$\delta_F\big|_\Delta = \det(M_F|_\Delta) = \det JF = c.$$

□

3.2 The idempotent

Let $\bar{\delta}_F := q_F(\delta_F)$ be the class of δ_F in C_F, and put

$$e_F := c^{-1}\bar{\delta}_F, \qquad p_F := 1 - e_F = 1 - c^{-1}\bar{\delta}_F.$$

The element e_F is the diagonal projector, while p_F is the off-diagonal collision projector.

Theorem 3.2 (Planar secant projector)**.** *With the notation above,*

$$e_F^2 = e_F, \qquad p_F^2 = p_F, \qquad e_F p_F = 0, \qquad e_F + p_F = 1.$$

Moreover,

$$\operatorname{Obs}(F) = C_F p_F, \qquad \operatorname{Ann}_{C_F}(\operatorname{Obs}(F)) = C_F e_F,$$

and p_F is the unique idempotent $u \in C_F$ satisfying $\operatorname{Obs}(F) = C_F u$*. Hence*

$$C_F \simeq A \times C_F p_F.$$

In particular,

$$p_F = 0 \quad \Longleftrightarrow \quad \operatorname{Obs}(F) = 0.$$

Proof. Proposition 3.1 gives

$$\bar{\delta}_F \operatorname{Obs}(F) = 0, \qquad \bar{\delta}_F - c \in \operatorname{Obs}(F).$$

Equivalently,

$$e_F \operatorname{Obs}(F) = 0, \qquad e_F - 1 \in \operatorname{Obs}(F).$$

Hence $e_F(e_F - 1) = 0$, so $e_F^2 = e_F$; the corresponding identities for $p_F = 1 - e_F$ follow.

The second relation says $p_F \in \operatorname{Obs}(F)$. If $j \in \operatorname{Obs}(F)$, then

$$j = (e_F + p_F)j = p_F j,$$

so $\operatorname{Obs}(F) = C_F p_F$. An element annihilates $C_F p_F$ exactly when its p_F-part is zero, giving

$$\operatorname{Ann}_{C_F}(\operatorname{Obs}(F)) = C_F e_F.$$

For uniqueness, let $u^2 = u$ and suppose $C_F u = C_F p_F$. Since $u \in C_F p_F$, one has $up_F = u$; since $p_F \in C_F u$, one has $up_F = p_F$. Thus $u = p_F$.

Complementary idempotents give

$$C_F \xrightarrow{\sim} C_F e_F \times C_F p_F, \qquad a \longmapsto (ae_F, ap_F).$$

Diagonal evaluation restricts to an isomorphism $C_F e_F \simeq A$: if $a = \bar{\mu}_F(r)$, then $a = \bar{\mu}_F(re_F)$, while its kernel is $C_F e_F \cap \operatorname{Obs}(F) = 0$. Thus $C_F \simeq A \times C_F p_F$. Finally, $\operatorname{Obs}(F) = C_F p_F$ gives

$$p_F = 0 \iff \operatorname{Obs}(F) = 0.$$

□

Abstractly, the existence of an idempotent splitting is expected: a Keller map is unramified, so its diagonal in the collision scheme is open, while a morphism between affine schemes has closed diagonal [The26e, The26d]. The theorem supplies the explicit Bézoutian representative and identifies the ideal it generates. The uniqueness just proved shows that the resulting p_F is independent of the ordering used to write the divided differences.

4 Quadratic generic collisions

We now work at the generic point. The essential field-theoretic fact is the degree-two marked-root decomposition.

4.1 The quadratic self-tensor product

Let L/K be finite separable of degree two. Paper I's marked-root tensor decomposition applies to a primitive generator α: it splits the marked root from the residual polynomial

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \qquad \text{in } L[T].$$

Here h_α has degree one, so its quotient algebra is another copy of L. Thus the general result of Paper I [vdV26] specializes as follows.

Proposition 4.1 (Quadratic marked-root decomposition)**.** *There is an L-algebra isomorphism*

$$\Psi_\alpha \colon L \otimes_K L \xrightarrow{\sim} L \times L$$

under which tensor multiplication

$$\mu_L \colon L \otimes_K L \longrightarrow L$$

is first projection. Consequently,

$$\Psi_\alpha\big(\ker(\mu_L)\big) = 0 \times L \neq 0.$$

4.2 Nonvanishing of the affine obstruction

Combining the quoted generic collision bridge with Proposition 4.1 transports the nonzero second factor back to the affine obstruction.

Theorem 4.2 (Quadratic collision nonvanishing)**.** *Let $F \colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}$ be a polynomial map whose induced extension L/K is finite separable of degree two. Then*

$$\mathrm{Obs}(F) \neq 0.$$

Equivalently,

$$I_R(F) \subsetneq I_\Delta.$$

Proof. Flatness of $B \to K$ gives a canonical identification

$$K \otimes_B \mathrm{Obs}(F) \simeq \ker(1 \otimes \bar{\mu}_F).$$

Under the compatible quadratic product, the right-hand side is

$$\ker\big(L \times L \xrightarrow{\mathrm{pr}_1} L\big) = 0 \times L,$$

which is nonzero. Therefore $K \otimes_B \mathrm{Obs}(F) \neq 0$, and hence $\mathrm{Obs}(F) \neq 0$. Since $\mathrm{Obs}(F) = I_\Delta / I_R(F)$, this is equivalent to strict containment. □

This theorem is independent of the Keller condition: generic degree two already forces a residual generic collision. For a planar Keller map, the secant projector makes that factor explicit.

Corollary 4.3 (Nonzero planar projector in generic degree two)**.** *Let*

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF = c \in \mathbb{C}^\times.$$

If $d_F = 2$, then

$$p_F \neq 0.$$

Proof. The Keller condition makes L/K finite separable. Theorem 4.2 gives $\mathrm{Obs}(F) \neq 0$, and Theorem 3.2 gives $\mathrm{Obs}(F) = C_F p_F$. Therefore $p_F \neq 0$. □

5 Galois rigidity and the collision contradiction

We now use the classical Galois case of the Jacobian conjecture.

Theorem 5.1 (Classical Keller–Galois rigidity)**.** *Let*

$$F\colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}$$

be a Keller map. If the induced finite separable extension L/K is normal, equivalently Galois, then F is a polynomial automorphism.

This is the classical normal/Galois case [Cam73, Raz79, Wri81]; we use it as a literature theorem and do not reprove it here.

Every separable extension of degree two is normal: if the minimal polynomial of a primitive generator has one root in L, its other root also lies in L. Thus Theorem 5.1 applies automatically on the generic-degree-two Keller locus.

Proposition 5.2 (Rigidity forces collision vanishing)**.** *Let*

$$F = (P, Q)\colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF = c \in \mathbb{C}^\times.$$

If $d_F = 2$, then Keller–Galois rigidity gives

$$\mathrm{Obs}(F) = 0, \qquad p_F = 0.$$

Proof. The extension L/K is finite separable of degree two, hence Galois. Theorem 5.1 makes F a polynomial automorphism. Theorem 2.1 then gives

$$I_R(F) = I_\Delta, \qquad \mathrm{Obs}(F) = 0.$$

Finally, Theorem 3.2 gives $p_F = 0$. □

Theorem 5.3 (Exclusion of generic degree two)**.** *No planar Keller map has generic degree two. Equivalently, every polynomial map*

$$F = (P, Q)\colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^\times,$$

satisfies

$$d_F \neq 2.$$

Proof. Suppose that $d_F = 2$. Corollary 4.3, which reads the residual factor in

$$K \otimes_B C_F \simeq_K L \times L,$$

gives $p_F \neq 0$. Proposition 5.2 gives $p_F = 0$. This is a contradiction. □

Equivalently, the two halves of the proof say

$$d_F = 2 \Longrightarrow \mathrm{Obs}(F) \neq 0$$

by the quadratic generic collision product, while

$$F \text{ Keller and } L/K \text{ normal} \Longrightarrow \mathrm{Obs}(F) = 0$$

by Galois rigidity and automorphism detection. The secant construction expresses both statements as nonvanishing and vanishing of one canonical idempotent.

6 Normal-closure and target-boundary criteria

The same normal closure used in Paper I also describes the rigidity case. We give exact conditions for vanishing, first in terms of the marked field and then in terms of the target boundary. Throughout this section, $F = (P, Q)$ is a planar Keller map, with no generic-degree restriction. The criteria do not assert vanishing for every such map.

6.1 The marked normal closure

Let N be the normal closure of L/K, fix the embedding $\iota\colon L \hookrightarrow N$, and put

$$G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/\iota(L)).$$

These are the marked objects of Paper I.

Proposition 6.1 (Normal-closure criterion)**.** *For a planar Keller map, the following conditions are equivalent:*

$$\mathrm{Obs}(F) = 0 \iff L/K \text{ is normal} \iff N = \iota(L) \iff H = 1.$$

Equivalently, every conjugate source coordinate satisfies $g(\iota(x_i)) \in \iota(L)$ for $g \in G$ and $i = 1, 2$.

Proof. If $\mathrm{Obs}(F) = 0$, then $I_R(F) = I_\Delta$. Evaluating at the collision pair $(\iota(x), g(\iota(x)))$ gives $g(\iota(x_i)) = \iota(x_i)$ for every $g \in G$ and $i = 1, 2$. The conjugate source coordinates generate N over K, so $N = \iota(L)$. This equality is equivalent to normality of L/K and, by Galois correspondence, to $H = 1$. Conversely, normality gives automorphy by Theorem 5.1, hence $\mathrm{Obs}(F) = 0$. The same generation statement shows that containment of all conjugate source coordinates in $\iota(L)$ suffices. □

Normality of N/K is supplied by the construction; normality of L/K is the criterion. In the cubic nonnormal case of Paper I, $\iota(L) \subsetneq N$ and $H \neq 1$. Here the complementary condition $N = \iota(L)$, together with Keller rigidity, forces $L = K$.

6.2 The reduced target boundary

Inside the same marked closure, set

$$E = B[g(\iota(A)) : g \in G] \subseteq N.$$

Thus E is generated by all conjugate source algebras. It is the coordinate ring of the connected component of their fiber product selected by these embeddings: the tensor product is regular and étale over B, and its selected generic factor is N. In particular, E/B is étale and $\mathrm{Frac}(E) = N$.

The action of G on $\mathrm{Spec}\, E$ is free. Indeed, the scheme-theoretic fixed locus of each $g \neq 1$ is open by unramifiedness and closed by separatedness. Since g acts nontrivially on $\mathrm{Frac}(E) = N$, this locus misses the generic point and is empty by connectedness. The quotient $\mathrm{Spec}\, E \to \mathrm{Spec}\, E^G$ is therefore a finite étale G-torsor [The26g]. Consequently $\mathrm{Spec}\, E^G \to \mathrm{Spec}\, B$ is étale, separated, and birational, hence an open immersion [The26c]. Write its image as U_F. The complement of this dense affine open has pure codimension one [The26a]; since $B = \mathbb{C}[P, Q]$ is factorial, we may write

$$U_F = D(s), \qquad \mathfrak{b}_F := I(\mathrm{Spec}\, B \setminus U_F) = (s), \qquad 0 \neq s \in B \text{ squarefree.}$$

For an empty complement take $s = 1$. The ideal $\mathfrak{b}_F$ depends only on U_F, not on its defining equation, and $E^G = B[1/s]$. It is a target-base boundary ideal, distinct from the collision obstruction $\mathrm{Obs}(F) \subseteq C_F$.

Proposition 6.2 (Source-regularity criterion)**.** *Identify A, B, and K with their marked images in N. Then*

$$A \cap K = B, \qquad E^G \cap A = B,$$

and

$$E^G \subseteq A \iff E^G = B \iff \mathfrak{b}_F = B.$$

Proof. The first identity is Wang's intersection theorem [Wan80, Theorem 41(i)]. We recall the divisibility argument used in the formalization. An irreducible $p \in B$ remains a nonunit in A, since the units of both polynomial rings are the nonzero constants. Choose an irreducible factor $q \in A$ of p. Flatness and going-down give height at most one for $(q) \cap B$; as this prime contains (p), it equals (p). Thus divisibility by p between elements of B is reflected from A to B. Factorization and cancellation reflect arbitrary divisibility. Clearing a fraction in $A \cap K$ now gives $A \cap K = B$. Since $E^G \subseteq N^G = K$, the second identity follows. It gives the first equivalence, and the open immersion $\operatorname{Spec} E^G \hookrightarrow \operatorname{Spec} B$ gives the second. □

These conditions are also equivalent to $\mathrm{Obs}(F) = 0$. Indeed, if $\mathfrak{b}_F = B$, then E is finite over B. Since B is Noetherian, the B-submodule $\iota(A) \subseteq E$ is also finite. Thus F is finite étale; the associated covering of the simply connected space $\mathbb{C}^2$ has one sheet, so F is a polynomial automorphism and $\mathrm{Obs}(F) = 0$. Conversely, the coordinate equalities in the proof of Proposition 6.1 give $E \subseteq \iota(A)$ when $\mathrm{Obs}(F) = 0$, and Proposition 6.2 gives $\mathfrak{b}_F = B$. This reverse implication uses the collision equations directly, without Ax–Grothendieck. The geometric implication from boundary emptiness uses finite-étale rigidity, separately from the Galois-rigidity argument above.

6.3 A finite first-order criterion

Let D_P, D_Q denote the extensions of ∂_P, ∂_Q from B to K [The26f], and set

$$M_0 = s^{-1}B, \qquad M_1 = M_0 + B\,D_P(1/s) + B\,D_Q(1/s) \subseteq K.$$

The module M_0 has one generator, M_1 has at most three, and M_1/M_0 has at most two. All finiteness statements here are over B, not finite-dimensionality over $\mathbb{C}$.

Proposition 6.3 (Finite boundary criterion)**.** *For every $0 \neq s \in B$, the following conditions are equivalent:*

$$M_1 = M_0;$$
$$D_P(M_0) \subseteq M_0, \qquad D_Q(M_0) \subseteq M_0;$$
$$\partial_P s, \partial_Q s \in (s);$$
$$s \in B^\times.$$

For the reduced boundary equation above, these conditions say $U_F = \operatorname{Spec} B$.

Proof. Since each target derivative preserves B, Leibniz reduces preservation of M_0 to its generator. For $j = P, Q$,

$$D_j(1/s) = -\frac{\partial_j s}{s^2} \in s^{-1}B \iff \partial_j s \in (s).$$

If s is nonconstant, every nonzero partial derivative has smaller total degree than s and cannot be a multiple of it. Both partials would therefore have to vanish, forcing s to be constant in characteristic zero, a contradiction. Nonzero constants satisfy all the displayed conditions. □

The same test has an intrinsic boundary formulation. The conormal map [The26b]

$$\mathfrak{b}_F/\mathfrak{b}_F^2 \longrightarrow \Omega_{B/\mathbb{C}} \otimes_B B/\mathfrak{b}_F \simeq (B/\mathfrak{b}_F)^2$$

sends the generator $s \bmod s^2$ to

$$(\partial_P s \bmod s,\ \partial_Q s \bmod s).$$

Its vanishing is equivalent to the criterion above. The intrinsic conormal criterion does not require a choice of the generator s. It uses the existing conormal module of the boundary, not a replacement for $\mathrm{Obs}(F)$.

The first-order test is another characterization of the same empty boundary. Neither field-level extension of the Keller frame nor preservation of the localized algebra $B[1/s]$ proves it: differentiation may send $1/s$ to $-\partial_j s/s^2$.

7 Lean formalization status

Lean correspondence. The theorem spine is formalized in Lean 4.28.0 and Mathlib v4.28.0 [dMU21, The20]. Shared collision objects and generic base change are inherited from `CollisionIdeals/General/Collision` and `CollisionIdeals/General/GenericFiber`. The quadratic collision decomposition is assembled in `CollisionIdeals/General/GenericFiber/QuadraticCollision.lean`; the explicit secant projector and degree-two contradiction are in `CollisionIdeals/Planar/ExplicitSecant.lean` and `CollisionIdeals/Planar/GenericDegreeTwo/Main.lean`.
The two Palomar-facing declarations are `CollisionIdeals.Palomar.PaperTwo.explicitPlanarSecantProjector` and `CollisionIdeals.Palomar.PaperTwo.quadraticPlanarCollisionRigidity`. They are checked by `lake build CollisionIdeals.Palomar.PaperTwo.Solution`.
The degree-two exclusion takes only `ComplexKellerGaloisRigidity 2` as an external mathematical input. This proposition represents Theorem 5.1 and is passed explicitly, neither installed as a Lean axiom nor proved here. Normalization, ramification, boundary, purity, and finite-étale interfaces do not enter that proof.
The marked normal-closure criterion is packaged in `CollisionIdeals/Planar/Research/Vanishing/GaloisDescent.lean`. Its vanishing-to-normality implication follows directly from the collision equations via `CollisionIdeals/General/Galois/CollisionVanishing.lean`; only the converse uses Galois rigidity. The general automorphism-detection theorem quoted from Paper I still retains its Ax–Grothendieck input.
The actual fixed algebra, target open, and reduced boundary are constructed in `CollisionIdeals/General/Keller`. Source intersection and source regularity use these same embedded rings.
The file `CollisionIdeals/Planar/Research/Vanishing/BoundaryCriterion.lean` packages obstruction vanishing equivalently as empty target boundary, fixed-source regularity, or vanishing of the two conormal components. Only the boundary-to-obstruction direction takes `BranchPurityA2` and `AffinePlaneFiniteEtaleRigidity`; neither Ax–Grothendieck nor Keller–Galois rigidity enters this characterization.
The principal boundary presentation remains unformalized; the intrinsic conormal criterion does not require it. No universal vanishing theorem is claimed.

8 Conclusion

For a planar Keller map, the divided-difference determinant gives

$$C_F \simeq A \times C_F p_F, \qquad \mathrm{Obs}(F) = C_F p_F,$$

where p_F is the explicit off-diagonal collision projector. Generic degree two gives a nonzero residual factor and hence $p_F \neq 0$. The quadratic extension is nevertheless Galois, so Keller–Galois rigidity forces polynomial automorphy and $p_F = 0$. This incompatibility excludes generic degree two.

The remaining possible counterexamples to $JC(2)$ have nonnormal function-field extensions of generic degree at least three. The degree-two case itself is complete once the classical rigidity theorem is taken as input. Beyond that case, the marked normal closure gives the exact criterion $\mathrm{Obs}(F) = 0$ if and only if $N = \iota(L)$. The target-boundary formulation is $E^G \subseteq A \iff E^G = B \iff \mathfrak{b}_F = B$, with the conormal test as an equivalent first-order condition. These characterizations do not assert that every planar Keller map satisfies them.

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